

Inbound REST Integrations

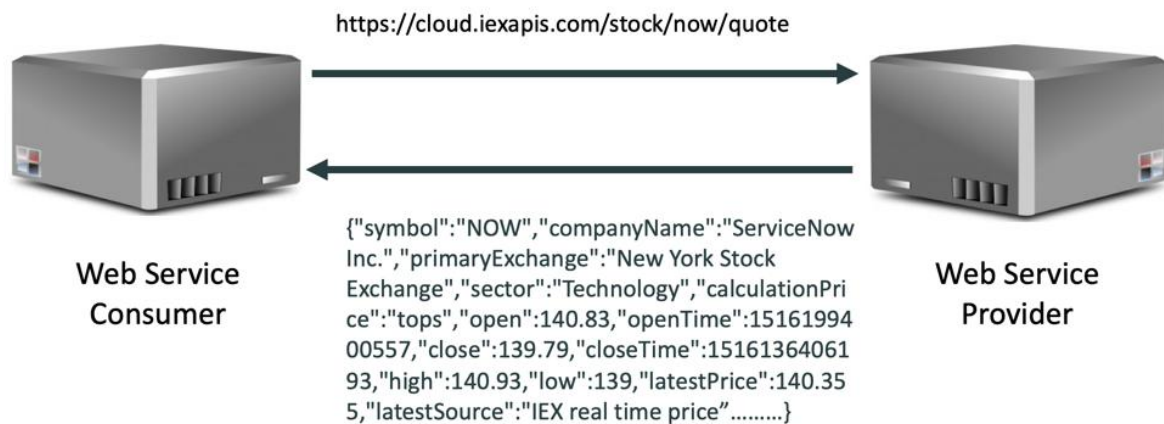
Inbound REST Integrations in ServiceNow Objectives

In these docs, you will:

- Explore the **REST API Explorer** and **Table API** to interact with ServiceNow table records, covering:
 - Path parameters
 - Query parameters
 - Headers
 - HTTP Status codes
 - Response bodies
- Set up a **ServiceNow user** for inbound REST requests
- Define **Cross-Origin Resource Sharing (CORS) rules** to control allowed HTTP methods from a resource
- Restrict web service access to specific tables
- Develop code samples for seamless integration of third-party applications with ServiceNow

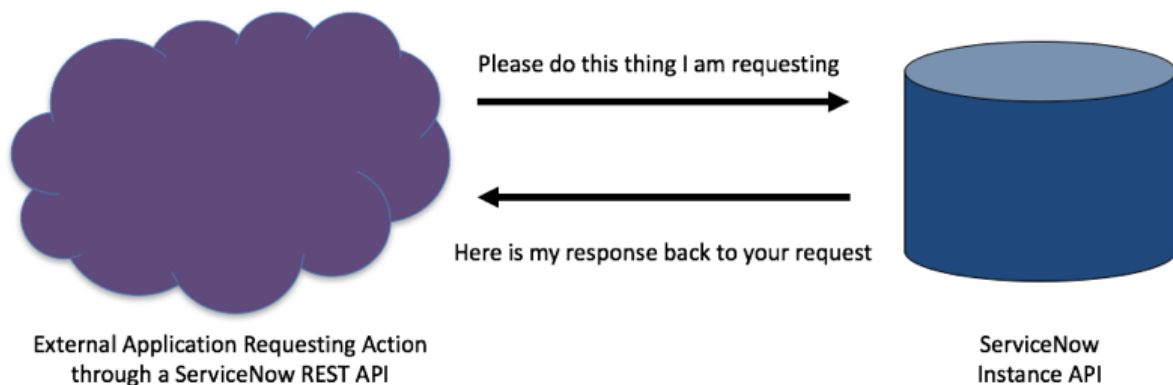
ServiceNow as a Web Service Provider

ServiceNow acts as a **web service provider**, enabling applications to connect and exchange data over a network. The **consumer (client)** sends requests, and the **provider (server)** processes them, returning a **status code** and **response body**. The client then extracts relevant data and takes action accordingly.



In an inbound request, a third-party application requests an action through a ServiceNow API. Example ServiceNow APIs include:

- **Table API:** Create, read, update, and delete records from a table
- **Attachment API:** Upload and query file attachments
- **Email API:** Send and receive email messages using REST



Introduction to the REST API Explorer

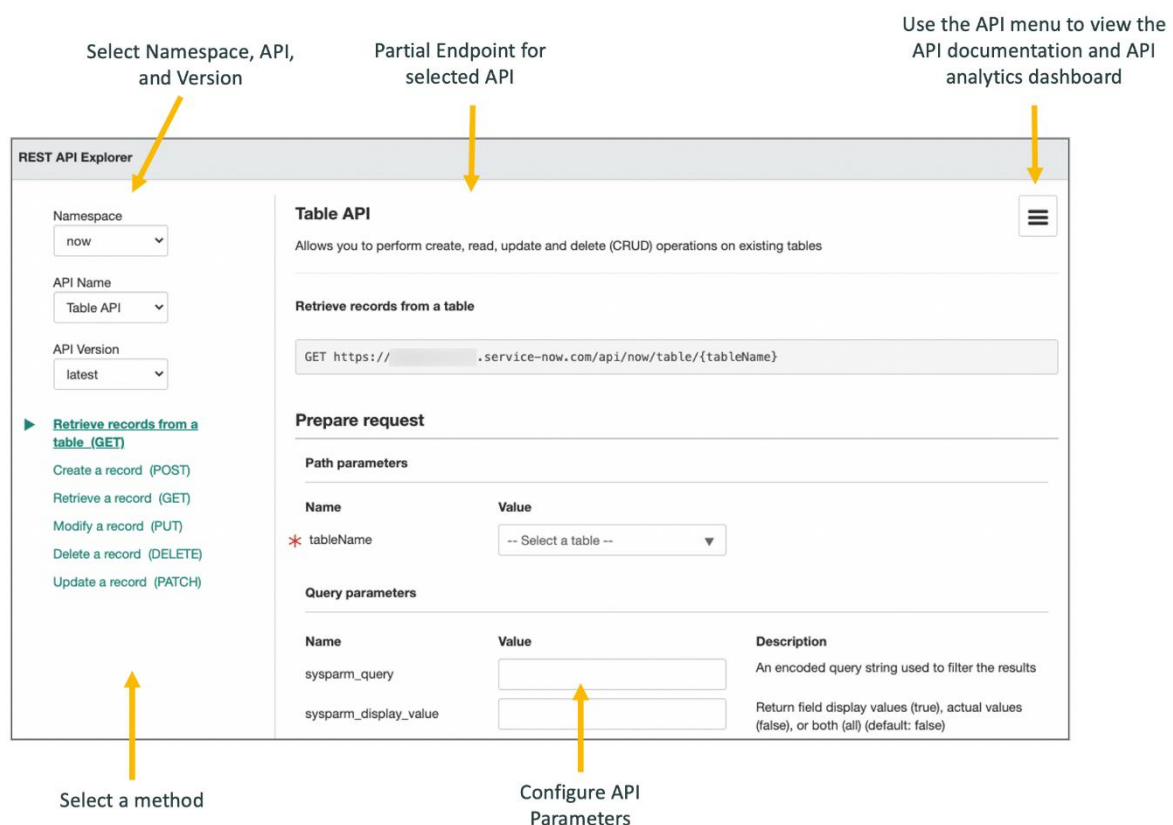
To integrate with a web service, developers must understand:

- **Endpoints**
- **Methods**
- **Variables**

ServiceNow's **REST API Explorer** allows users to construct and test API requests within a ServiceNow instance. It is accessible to those with the **rest_api_explorer** or **admin** role. To open it, navigate to **System Web Services > REST > REST API Explorer** via the **All** menu.

The **REST API Explorer** includes:

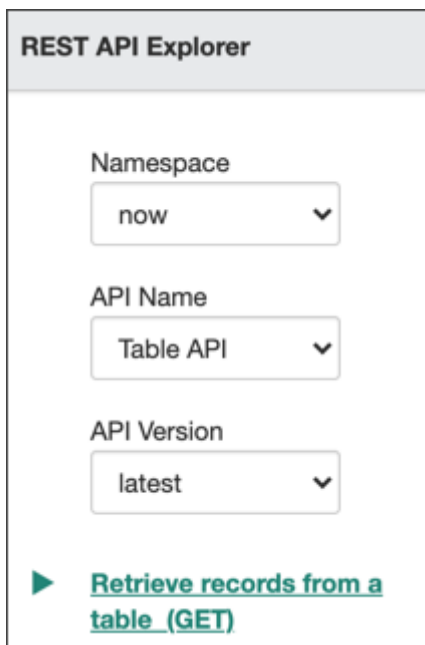
- A pane to select **Namespace, API Name, API Version, and REST method**
- A pane to **view and configure the endpoint**
- A menu for **API documentation** and an **API analytics dashboard**
- A section to **test the endpoint** (not shown in the image)



Selecting an API

The fields in the *Prepare request* section of the REST API Explorer form are determined by which *Namespace*, *API Name*, *API Version*, and *REST method* is selected.

- **Namespace:** Select the web service scope
 - **global:** Globally scoped APIs
 - **now:** REST APIs provided by ServiceNow
 - **private_scope_name:** Namespace format used for APIs (scripted web services) in privately-scoped applications
- **API Name:** Select an API to configure and test in the REST API Explorer
- **API Version:** Select a specific API version or choose latest
- **Method:** Select from the list of available REST methods based on the *Namespace*, *API Name*, and *API Version*. The arrowhead indicates the selected method.



The screenshot shows the 'REST API Explorer' interface. It has three dropdown menus: 'Namespace' with 'now' selected, 'API Name' with 'Table API' selected, and 'API Version' with 'latest' selected. Below these is a green arrow icon followed by the text 'Retrieve records from a table (GET)'.

- For the *Aggregate API*, only one method is available.

The REST API Explorer displays information about the selection. For more information, select the **API documentation** menu item from the *REST API Explorer* menu.



The screenshot shows the 'Aggregate API' page. It has a title 'Aggregate API' and a description 'Allows you to compute aggregate statistics about existing table and column data'. Below the description is a section 'Retrieve statistical calculations for a table'. At the bottom, there is a text box containing the REST endpoint 'GET https://(redacted).service-now.com/api/now/stats/{tableName}'. On the right side, there is a menu icon (three horizontal lines) and a dropdown menu with three items: 'Share link', 'API documentation' (which is highlighted), and 'API analytics'.

Request Parameters

Request Parameters consist of:

- *Path parameters*
- *Query parameters*
- *Request headers*

Path Parameters

Path parameters vary based on the endpoint URL and are enclosed in **curly braces** {}. When a request is sent, the values entered in the path parameter field replace the placeholders in the URL.

Aggregate API

Allows you to compute aggregate statistics about existing table and column data

Retrieve statistical calculations for a table

GET https://[redacted].service-now.com/api/now/stats/{tableName}

Prepare request

Path parameters

Name	Value
* tableName	Incident (incident) ▼



Query Parameters

The **REST API Explorer** adds request parameters to the endpoint URL when sending a request. **Query parameters** are specific to the selected API method.

Query parameters

Name	Value	Description
sysparm_query		An encoded query string
sysparm_avg_fields		A comma-separated list of fields for which to calculate the average value
sysparm_count		Calculate the number of records (default: false)
sysparm_min_fields		A comma-separated list of fields for which to calculate the minimum value
sysparm_max_fields		A comma-separated list of fields for which to calculate the maximum value
sysparm_sum_fields		A comma-separated list of fields for which to calculate the sum of the values
sysparm_group_by		A comma-separated list of fields to group results by
sysparm_order_by		A comma-separated list of fields to order results by
sysparm_having		An additional query allowing you to filter the data based on an aggregate operation
sysparm_display_value		Return the display value (true), actual value (false), or both (all) in grouped results (default: false)
sysparm_query_category		Name of the query category (read replica category) to use for queries

Add query parameter

The API displays a default set of **query parameters**. To add more, click **Add query parameter**. For a full list and details, access the **API documentation** from the **REST API Explorer menu**.

Request Headers

Request headers define the format of the Request and Response.

Request headers

Name	Value	Description
Request format	✓ application/json	Format of REST request body
Response format	application/xml text/xml	Format of REST response body
Authorization	Send as me	Send the request as the current user. To send the request with another user's credentials use the provided code samples, such as cURL.

Add header

Use the **Add header** button to add additional headers to the request. For the ServiceNow APIs, two useful additional header parameters are:

- ***X-WantSessionNotificationMessages***: Set to *true* to return notifications that have not already been consumed for the existing session.
- ***X-WantSessionDebugMessages***: Enable Session Debug and set the header value to *true* to return session debug logs.

Request headers

Name	Value
Request format	application/json
Response format	application/json
Authorization	Send as me

X-WantSessionDebugMessages

true

More About Query Parameters

Query parameters may contain different types of values, such as:

- Encoded queries
- Field name(s)
- true/false/all

Encoded Queries

Certain query parameters, like **sysparm_query** and **sysparm_having**, accept **encoded queries**, which are difficult to create manually and not documented. Let **ServiceNow** generate them instead:

1. In the **ServiceNow browser**, open the table list via the **All menu**.
 - If no module exists, type .list in the **Filter navigator**.
2. Use the **Filter** to build the query condition.

Incidents Number Search

Run Save... AND OR Add Sort

All of these conditions must be met

Active	is	true	AND OR X
Created	on	Last 12 months	AND OR X
Priority	is one of	1 - Critical 2 - High 3 - Moderate 4 - Low	AND OR X

Click the **Run** button to execute the query. Right-click the breadcrumbs and select **Copy query**.

NOTE: The copied query includes the condition right-clicked on and all conditions to the left. To copy the entire query, right-click **<the condition farthest to the right>**.

Incidents Number Search

All > Active = true > Created on Last 12 months > Priority in (1 - Critical, 2 - High, 3 - Moderate, 4 - Low)

Number	Opened	Short description	Caller
INC0000055	2021-11-08 20:47:23	SAP Sales app is not accessible	Carol Coogman

Return to the REST API Explorer and paste the encoded query into the query parameter.

Query parameters		
Name	Value	Description
sysparm_query	active=true^sys_created_onONLast 12 mont	An encoded query string used to filter the results

Field Names

Some query parameters, such as *sysparm_group_by*, accept a field name or a comma separated list of field names. Developers must pass the field *name* not the field *label*.

To select fields from a slushbucket, click the **Edit** button (



) on the field.

Configuring sysparm_group_by

Available

- Company
- Reassignment count
- Activity due
- Assigned to**
- Severity
- Additional comments
- SLA due
- Approval
- Comments and Work notes
- Due Date
- Updates
- Reopen count
- Tags
- Escalation
- Upon approval
- Correlation ID
- Location
- Category

>

<

Selected

^

v

Cancel

Save

sysparm_group_by

assigned_to

A comma-separated list of fields to group results by

Dot-walking is allowed in field names, for example, *caller_id.title*.

true/false/all

When a REST call returns a reference field from a table, developers can select the format for the returned value.

- Set the value to **true** to return the display value
- Set the value to **false** to return the *sys_id*
- Set the value to **all** to return both the display value and the *sys_id*

sysparm_display_value

all

Return the display value (true), actual value (false), or both (all) in grouped results (default: false)

Testing

After configuring the REST method, click the **Send** button to send the request to the API.

Request headers

Name	Value	Description
Request format	application/json	Format of REST request body
Response format	application/json	Format of REST response body
Authorization	Send as me	Send the request as the current user. To send the request with another user's credentials use the provided code samples, such as cURL.

Add header

Send

The REST API Explorer responds as if the request came from a third party application:

- *Request*
- *Response*
- *Response Body*

Request

HTTP Method / URI

GET https://[REDACTED].service-now.com/api/now/stats/incident?sysparm_query=active%3Dtrue%5Esys_created_onONLast%2012%20m

Headers

Accept

application/json

Content-Type

application/json

X-UserToken

c4eecfbadb3aa340a40afd131d9619b61327ec7810cfa5d3caa073e71459825e560b955a

Response

Status code

200 OK

Execution time (ms)

137

Headers

Cache-Control

no-cache,no-store,must-revalidate,max-age=-1

Content-Encoding

gzip

Content-Type

application/json;charset=UTF-8

Date

Tue, [REDACTED] 16:29:02 GMT

Expires

0

Pragma

no-store,no-cache

Server

ServiceNow

Strict-Transport-Security

max-age=63072000; includeSubDomains

Transfer-Encoding

chunked

X-Is-Logged-In

true

X-Transaction-Id

610b97b2db7a

Response Body

```
{
  "result": [
    {
      "stats": {
        "count": "3"
      },
      "groupby_fields": [
        {
          "field": "assigned_to",
```

ServiceNow API Request

The REST API Explorer constructs the request to send to the ServiceNow API using the settings configured by the developer.

Path Parameters in the Request

The *Request* section displays the HTTP Method / URI to send to the ServiceNow web service. The method is from the selected API. The path parameter values are set when configuring the request.

Aggregate API
Allows you to compute aggregate statistics about existing table and column data

Retrieve statistical calculations for a table

GET https://...service-now.com/api/now/stats/{tableName}

Prepare request

Path parameters

Name	Value
* tableName	Incident (incident)

Request

HTTP Method / URI

GET https://...service-now.com/api/now/stats/incident?sysparm_query=active%3Dtrue%26sys_created_on%3C%202012-12-31T23:59:59.999Z%40javascript%3A...

Headers

Accept	application/json
Content-Type	application/json
X-UserToken	c4eecfbadb3aa340a40afd131d9619b61327ec7810cfa5d3caa073e71459825e560b955a

Query Parameters in the Request

The query parameters are added to the URI. In the example shown, the URI is truncated due to space limitations. To see the complete URI in the REST API Explorer, scroll horizontally.

Query parameters

Name	Value
sysparm_query	active=true^sys_created_onONLast 12 mont
sysparm_avg_fields	
sysparm_count	true
sysparm_min_fields	
sysparm_max_fields	
sysparm_sum_fields	
sysparm_group_by	assigned_to
sysparm_order_by	
sysparm_having	
sysparm_display_value	all
sysparm_query_category	

Request

HTTP Method / URI	GET https://...service-now.com/api/now/state/incident?sysparm_query=active%3Dtrue%5Esys_created_onONLast%2012%20months%40javas
Headers	
Accept	application/json
Content-Type	application/json
X-UserToken	c4eecfbadb3aa340a40afd131d9619b61327ec7810cfa5d3caa073e71459825e560b955a

Headers in the Request

REST headers are the meta-data associated with an API request and response. The Request Header settings appear in the request.

The diagram illustrates the flow from the 'Request headers' configuration panel to the 'Request' details view. An orange arrow points from the 'Request headers' section to the 'Headers' section of the 'Request' view.

Request headers configuration:

Name	Value
Request format	application/json
Response format	application/json
Authorization	Send as me

Request details:

HTTP Method / URI: GET https://[redacted].service-now.com/api/now/stats/incident?sysparm_query=active%3Dtrue%5Esys_created_onONLast%2012%20months%40javasc

Headers:

Accept	application/json
Content-Type	application/json
X-UserToken	c4eecfbadb3aa340a40afd131d9619b61327ec7810cfa5d3caa073e71459825e560b955a

ServiceNow API Response

The ServiceNow API Response consists of:

- HTTP status code
- Response headers
- Response body

HTTP Status Code

ServiceNow APIs return standard HTTP status codes. Generally speaking:

- **1xx**: Informational
- **2xx**: Success
- **3xx**: Redirection
- **4xx**: Client Error
- **5xx**: Server Error

Response	
Status code	200 OK

The HTTP status codes refer to the interaction with the REST service provider. The status codes do not tell anything about the requested data. The REST transaction request can complete successfully even if no data is returned.

Response

Status code200 OK

Execution time (ms)111

Headers

Cache-Control	no-cache,no-store,must-revalidate,max-age=-1
Content-Encoding	gzip
Content-Type	application/json;charset=UTF-8
Date	Tue, 08 Jan 2019 17:18:50 GMT
Expires	0
Pragma	no-store,no-cache
Server	ServiceNow
Strict-Transport-Security	max-age=63072000; includeSubDomains
Transfer-Encoding	chunked
X-Is-Logged-In	true
X-Transaction-Id	8f66e37adb7a

Response Body

```
{  "result": []}
```

Response Headers

The *Headers* section shows the returned headers and their values.

Headers	
Cache-Control	no-cache,no-store,must-revalidate,max-age=-1
Content-Encoding	gzip
Content-Type	application/json;charset=UTF-8
Date	Mon, 05 Feb 2018 23:17:45 GMT
Expires	0
Pragma	no-store,no-cache
Server	ServiceNow
Strict-Transport-Security	max-age=63072000; includeSubDomains
Transfer-Encoding	chunked
X-Is-Logged-In	true
X-Transaction-Id	03c815664f00

Response Body

The response body is the data object returned by the ServiceNow web service provider. The response body varies depending on the selected API. In the example, the *Aggregate API* returns the count of open incident records in the past year with a priority of *Critical* or *High*. The results are grouped by the user in the *Assigned to* field.

Response Body
<pre>{ "result": [{ "stats": { "count": "3" }, "groupby_fields": [{ "field": "assigned to", "value": "46d44a23a9fe19810012d100cca80666", "display_value": "Beth Anglin" }] }], { "stats": { "count": "1" } } }</pre>

Beth Anglin has three *Critical* or *High* priority incidents still open from the past year. Notice in this case both the value (*sys_id*) and *display_value* are included for the *assigned_to* field.

Data Types and Returned Values

- **Encrypted text:** The database value is encrypted, while the displayed value is unencrypted based on the user's encryption context.
 - **Reference fields:** The database value is a *sys_id* and the display value is the human readable name.
 - **Date fields:** The database value is in UTC format, while the display value is based on the user's time zone.
 - **Choice fields:** The database value may be a number, while the display value will be more descriptive.
 - **Currency fields:** When performing a REST request, returned currency values are converted to local currency based on the user's locale. When inserting data, no conversion is performed.
-

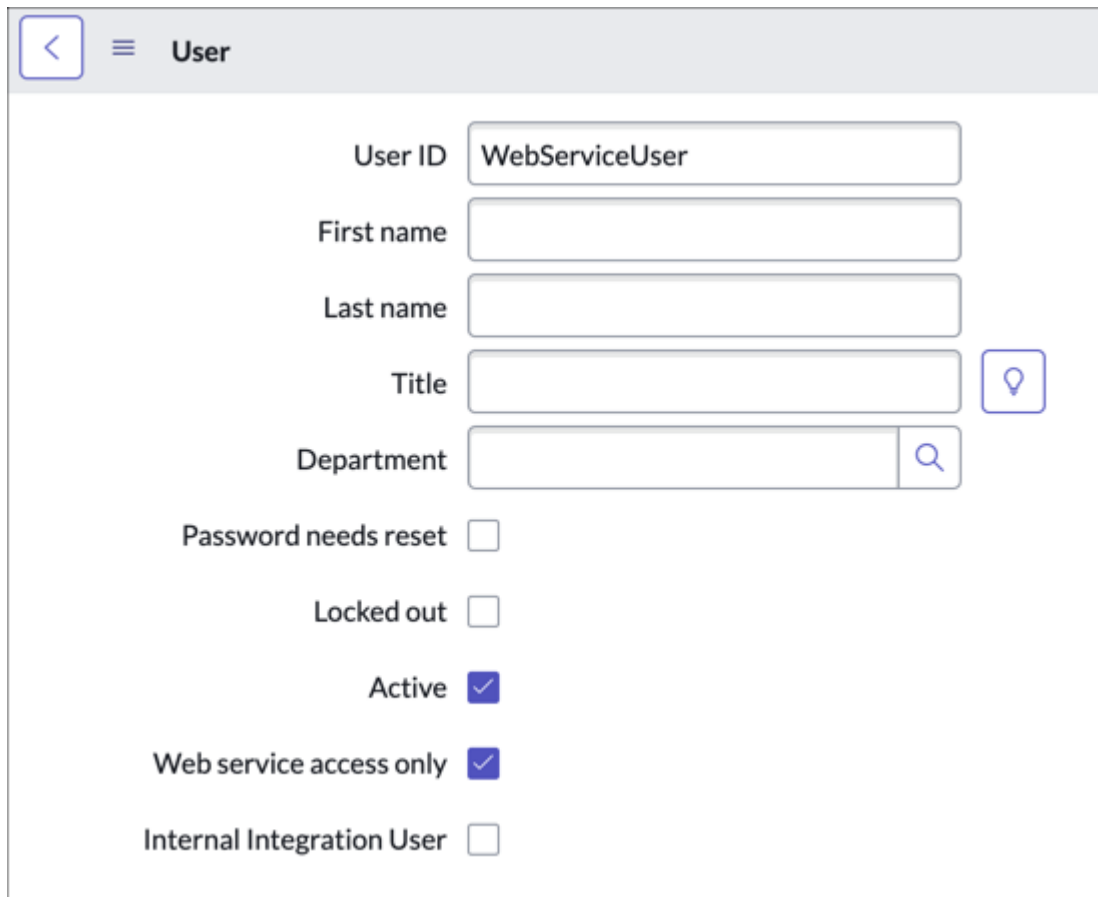
Adding Security to Inbound Requests

Strategies to add security to inbound API requests include:

- Create a user specifically for inbound requests
- Disallow web service access to tables
- Create CORS rules

Create an API Request User

Users with the *Web service access only* option set on their user record cannot log into the ServiceNow UI. This option allows the user credentials to be used only to authorize API connections. To set this option, open the user record for editing using the **User Administration > Users** module.



The screenshot shows the 'User' record form in ServiceNow. The header bar includes a back arrow, a menu icon, and the title 'User'. The form contains the following fields and options:

- User ID:** Text field containing 'WebServiceUser'.
- First name:** Empty text field.
- Last name:** Empty text field.
- Title:** Empty text field with a lightbulb icon to the right.
- Department:** Text field with a search icon to the right.
- Password needs reset:** ☐
- Locked out:** ☐
- Active:** ☒
- Web service access only:** ☒
- Internal Integration User:** ☐

The API request user must be granted the roles necessary to access the records requested by the API requests.

Disallow Web Service Access to Tables

Administrators can disable web service access to tables. On the table record, open the **Application Access** section and de-select the **Allow access to this table via web services** option. REST requests are not accepted for tables unless this option is selected (checked). To set this option, open the table record for editing using the **System Definition > Tables** module.

CORS Rules

Cross-Origin Resource Sharing (CORS) rules control which domains can access specific REST API endpoints. To create a CORS rule, use the *All* menu to open **System Web Services > REST > CORS Rules**.

In the example, the resource <https://www.test-cors.org> can only access the *Table API* using the *GET* method.

- **REST API:** The REST API the CORS rule applies to.
- **Domain:** The domain for the CORS rule. Specify the domain using an IP Address or a domain pattern.

- **Max age:** The number of seconds to cache the client session. After an initial CORS request, further requests from the same client within the specified time do not require a preflight message. If a value is not specified, the default value of 0 indicates that all requests require a preflight message.
- **HTTP Methods:** The methods allowed.
- **HTTP Headers:** A comma-separated list of HTTP headers to send in the response. Specified headers are added to the *Access-Control-Expose-Headers* header.

There are a number of requirements for specifying the domain including:

- Start with *http://* or *https://*
- Must be an IP address or domain pattern
- Can contain only one wildcard *

Code Samples

The REST API Explorer creates code samples for integrating with the ServiceNow APIs in several commonly used languages:

- *ServiceNow Script*
- *cURL*
- *Python*
- *Ruby*
- *JavaScript*
- *Perl*
- *Powershell*

To create the code sample, click the link in the REST API Explorer.

Code Samples

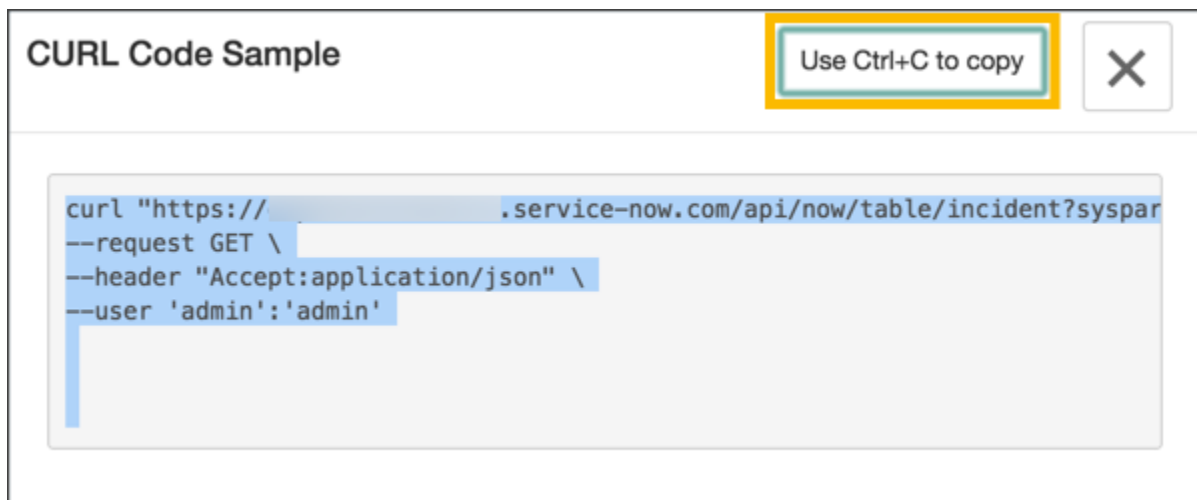
Use the provided code samples to send this request from commonly used languages.

[\[ServiceNow Script\]](#) [\[cURL\]](#) [\[Python\]](#) [\[Ruby\]](#) [\[JavaScript\]](#) [\[Perl\]](#) [\[Powershell\]](#)

To highlight the code sample for copying, click the **Select Snippet** button.



After highlighting the code sample, copy the code sample to the clipboard.



All code samples use fake credentials. Before using the script in the application to integrate with ServiceNow, update the code to use valid credentials.

Inbound Integrations Module Recap

Core concepts:

- Use the REST API Explorer to create and test inbound ServiceNow API requests
- Path parameters are part of the endpoint URL
- Query parameters determine which records, which data, and the data format returned in the response body
- The HTTP status code indicates the status of the transaction request and does not indicate any information about the returned data
- The response body format is set in the headers
- The REST API Explorer tests requests as the currently logged in user
- The *Allow access to this table via web services* option cannot be tested in the REST API Explorer
- Do not use the admin user in code integrations, create a web services only user instead
- Disable web service access to tables with sensitive data unless web service access is required
- CORS rules add security to APIs
 - CORS rules determine which cross-origin resources can access which methods
 - CORS rules cannot be tested in the REST API Explorer
- Code samples provide script stubs for integrating into ServiceNow from third party applications
 - Developers must update credentials in the code samples
 - Code samples are available in multiple standard languages used with integrations